Lab 2: Protocol Layers and Security Design

Network System Design | SE026 | 2026-27 Name: LOR HENG RITH

1. Record Your Observation

Item	Value
OS / Wireshark version	4.6.9
Capture filename	`Week2-Teaching-Trace.pcap`
Unfiltered packet count	**12**

2. DNS and Protocol Fields

Filter used: `dns`

Observation	Evidence and explanation
DNS query / response frame numbers	Query = **frame 1**, Response = **frame 2**
Query name and record type	`data.example.test`, type **A** (IPv4 address record)
Query source / destination IPv4	Query: `192.0.2.10` → `192.0.2.53`. Response: `192.0.2.53` → `192.0.2.10`
UDP source / destination port	Query: `53000` → `53`. Response: `53` → `53000`
Returned address	`198.51.100.20` (TTL 60 s)

Extra evidence: both frames share DNS transaction ID `0x1234`, which is how the response is matched to the query. Frame 1 has flags `0x0100` (standard query, recursion desired). Frame 2 has `0x8180` (response, recursion available, no error).

Explain the name-lookup protocol, the protocol carrying this exchange and the layer supplying IPv4 delivery.

- **DNS** is the name-lookup protocol (application layer). It maps the name `data.example.test` to an IPv4 address.
- **UDP** (transport layer) carries the DNS messages. The client uses an ephemeral port (53000) and the server listens on well-known port 53.
- **IPv4** (network layer) supplies host-to-host delivery between `192.0.2.10` and `192.0.2.53`.

Does this trace prove that every DNS exchange uses UDP? Explain the evidence limit.

No. The trace contains only one DNS exchange (frames 1–2), so it shows that *this* exchange used UDP, not that all DNS does. DNS can also run over TCP (for example when a response is too large for UDP, for zone

transfers, or with DNS-over-TLS). This capture has no such traffic, so it cannot rule it out. It is a single sample from one observation point.

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3. HTTP Request and Encapsulation

Filter used: `http`

Layer evidence	Your field and meaning
Ethernet source / destination MAC	Request (frame 6): src `02:00:00:00:00:10` (client) → dst `02:00:00:00:00:01` (next hop, i.e. the gateway)
IPv4 source / destination address	`192.0.2.10` (client) → `198.51.100.20` (web server)
TCP source / destination port	`51000` (ephemeral client port) → `80` (HTTP)
HTTP method / Host / resource	`GET` / `data.example.test` / `/dataset.json` (HTTP/1.1)
Response frame / status	**Frame 8**, `200 OK` (`Content-Type: application/json`, `Content-Length: 38`)

Follow > TCP Stream: which text is visible and which layers carry it.

Request (frame 6, client → server):

```
GET /dataset.json HTTP/1.1
Host: data.example.test
Connection: close
```

Response (frame 8, server → client):

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 38
Connection: close

{"dataset":"classroom_demo","rows":2}
```

All of this is readable as plain text because it is **plain HTTP with no encryption**. The text is the application-layer payload of a TCP segment. That segment sits in an IPv4 packet, which sits in an Ethernet frame. Anyone who can observe the link can read the method, host, path, headers and body.

Supporting TCP frames in the same stream:

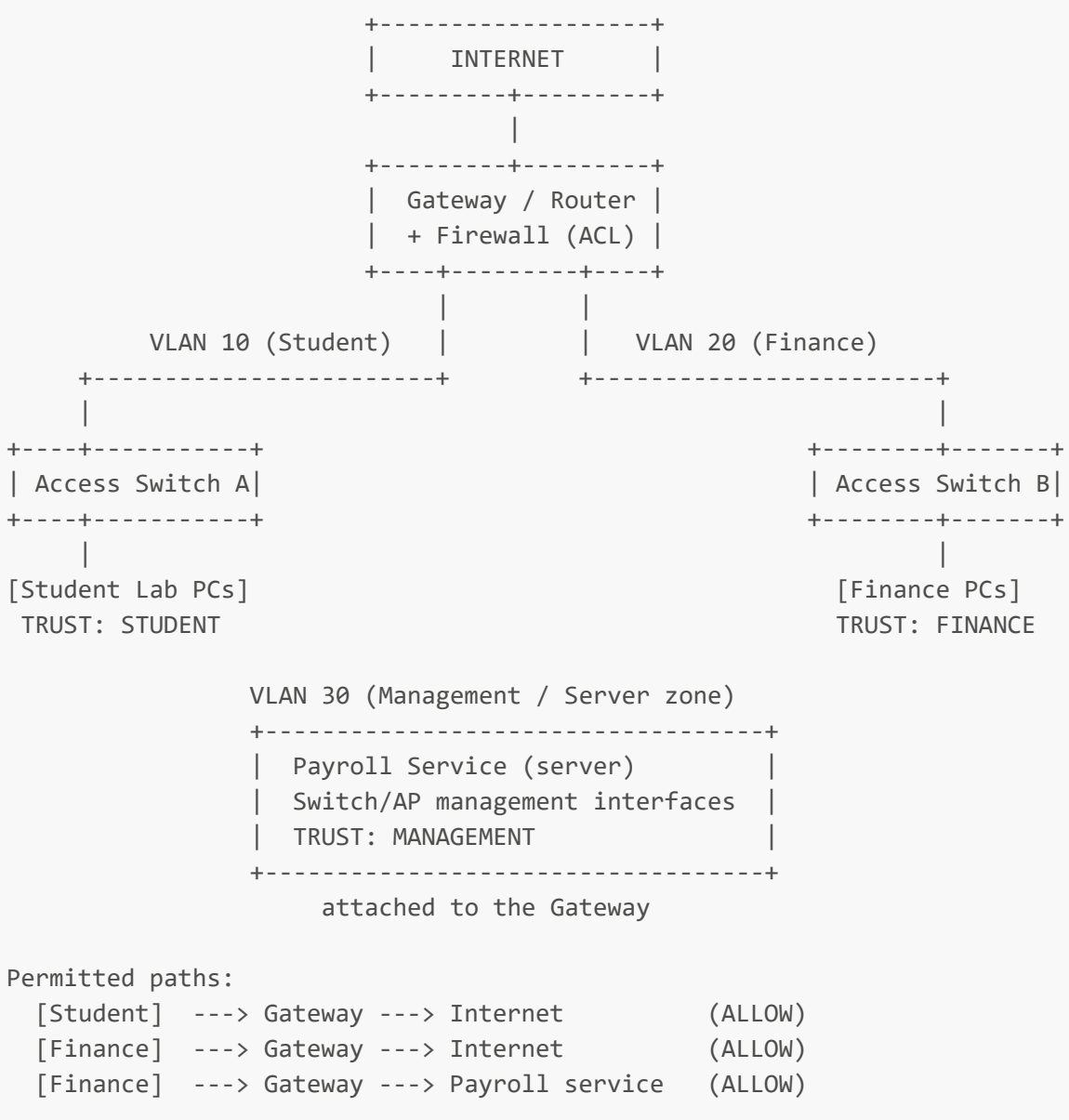
Frames	Purpose
3, 4, 5	Three-way handshake (SYN, SYN-ACK, ACK)
6	HTTP request (PSH, ACK)
7	Server ACK of the request
8	HTTP response (PSH, ACK)
9	Client ACK of the response
10, 11, 12	Connection close (FIN-ACK, FIN-ACK, ACK)

Note: the HTTP request is in frame 6 and the response in frame 8. The TCP handshake and close frames carry no HTTP text.

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4. Forwarding and Trust Boundaries

Design diagram



```
Restricted path:
  [Student]  -X-> Gateway -X-> Payroll service   (DENY)
```

Trust groups: **Student** (VLAN 10), **Finance** (VLAN 20), **Management** (VLAN 30: payroll service and device management).

Requirement / threat table

Requirement / threat	Proposed control	Verification / limit
Only finance users may reach the payroll service; students must be blocked	Separate VLANs per trust group plus a firewall/ACL rule at the gateway: permit Finance subnet → Payroll, deny Student subnet → Payroll (default deny)	From a student PC, try to connect to the payroll service (ping, TCP connect on its port) and confirm it fails. Repeat from a finance PC and confirm it works. Check the firewall hit counters/logs. **Limit:** tests only the paths tried; also needs a review of rule order and of any other route to the server.
Both groups need Internet access	Allow outbound Internet from Student and Finance VLANs through the gateway (with NAT)	Browse or ping an external address from both VLANs. **Limit:** shows reachability, not that the allowed traffic is safe.
Confidentiality threat: plain HTTP exposes content (frames 6 and 8 are readable by anyone on the path)	Use **HTTPS (TLS)** for web and payroll traffic, and disable or redirect plain HTTP	Capture again in Wireshark and confirm the HTTP text is no longer readable (shows TLS records instead). **Limit:** IP addresses, ports, and often the server name (SNI) and traffic volume remain visible; TLS does not hide metadata.

Forwarding explanation

When a packet is forwarded by a router (with no NAT, tunnel or proxy):

- **The Ethernet header changes on every link.** The router strips the incoming frame, decrements the IP TTL, looks up the next hop, and builds a new frame with a new source MAC (the router's outgoing interface) and a new destination MAC (the next hop or the final host on that link).
- **The IP source and destination addresses stay the same** end to end (here `192.0.2.10` → `198.51.100.20`).

In the trace, frame 6 on the client's link has dst MAC `02:00:00:00:00:01`, which is the gateway and not the web server. The server's identity is carried only in the IP destination `198.51.100.20`.

Conceptual prediction (not in the trace) for the next link: the same IP packet would appear in a new Ethernet frame with src MAC = the router's outgoing interface and dst MAC = the next-hop router (or the server). IP src/dst would still be `192.0.2.10` → `198.51.100.20`, with TTL reduced by 1.

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5. Reflection

1. Explain how an HTTP message fits inside TCP, IP and Ethernet. The HTTP request or response is the application data. TCP wraps it in a segment with source/destination ports and sequence numbers (here 51000 → 80). IP wraps the segment in a packet with source/destination IP addresses (192.0.2.10 → 198.51.100.20). Ethernet wraps the packet in a frame with source/destination MAC addresses for the local link. Each layer adds its own header, and the receiver removes them in reverse order.

2. Why can the destination MAC identify a different device from the remote server named by the IP destination? The MAC address is only valid on the local link and identifies the *next hop*, here the gateway (`02:00:00:00:00:01`). The IP destination names the final remote server (`198.51.100.20`), which is not on the same link. The router rewrites the MAC at each hop while the IP destination stays unchanged.

3. Which security goal does your proposed control address? VLAN separation plus the ACL/firewall rule addresses **access control / authorization** (and limits who can reach payroll, which also protects confidentiality and integrity of payroll data). HTTPS/TLS addresses **confidentiality** (and integrity) of data in transit.

4. What evidence is still needed before implementing the design?

- Real traffic measurements (who talks to what, and on which ports), since this trace is only a teaching example and not campus traffic.
- The actual building layout, switch/VLAN capabilities and gateway/firewall model.
- A confirmed list of users and devices per trust group, and the payroll service's address and ports.
- Whether any other services or legacy plain-HTTP apps depend on current access, plus a rollback and test plan.